Construction problems with compasses and a straightedge

Four basic constructions combined into triangles — and the rule that decides when it can be done at all.

LESSON 133–134 | 2 × 40 MIN | MATEMATIKA 7, §58 | S8 5.3

LESSON CLOCK

10' | 25' | 25' | 20' | 5'

The rules of the game	10 min
A triangle from three sides	25 min
The other two triangles	25 min
Perpendiculars and midpoints	20 min
Homework	5 min

LEARNING OBJECTIVES

- Construct a triangle from three sides, from two sides and the angle between, and from a side and two angles.
- Decide when a construction is impossible.
- Construct the perpendicular from a point to a line and the midpoint of a segment.
- Justify a construction with a congruence criterion.

TEXTBOOK

National	pp. 393–400
Cambridge	Stage 8, pp. 52–56
Workbook	Exercise 5.3

01 Explanation

The rules of the game

Only two tools are allowed: a straightedge, which draws a line through two points, and compasses, which draw a circle of a given radius. No measuring with the ruler's scale, and no protractor.

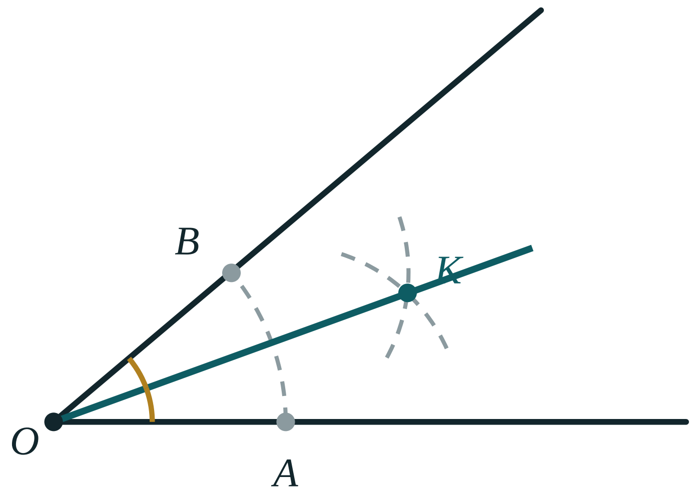
ALLOWED	NOT ALLOWED
a line through two points	measuring a length with the scale
a circle with a given centre and radius	measuring an angle with a protractor
marking where two drawn curves cross	sliding the ruler until it looks right

LEAVE EVERY ARC ON THE PAGE

The arcs are the proof that the construction was done and not estimated. Rubbing them out removes the working, and with it the marks.

A triangle from three sides

Given three lengths a, b, c:



Arcs meeting to fix the third vertex

STEP	WHAT TO DO
1	draw a segment $BC = a$
2	with centre B and radius c, draw an arc
3	with centre C and radius b, draw a second arc
4	call the crossing A and join AB and AC

Every triangle built this way is congruent to every other, by the third criterion — which is why the three sides determine the triangle completely.

THE ARCS MEET ONLY IF THE TRIANGLE INEQUALITY HOLDS

With $5, 8, 14$ the arcs fall short of each other and there is nothing to mark. The construction fails exactly when the triangle does not exist.

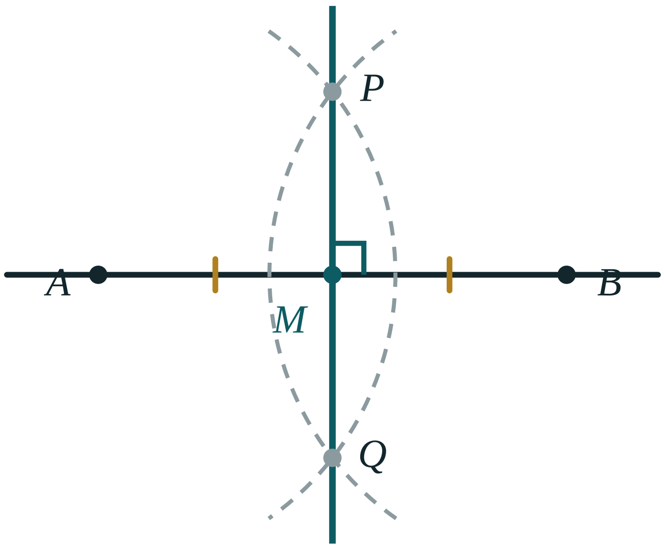
The other two triangles

GIVEN	CONSTRUCTION	JUSTIFIED BY
three sides	two arcs from the ends of the first side	the third criterion
two sides and the angle between	copy the angle, then mark the two lengths along its arms	the first criterion
a side and the two angles on it	copy each angle at one end of the side	the second criterion

THE THREE CRITERIA ARE THE THREE CONSTRUCTIONS

Each congruence criterion says which data fix a triangle; each construction is the recipe for building the triangle those data describe. The chapters answer each other.

Perpendiculars and midpoints



The perpendicular bisector of a segment

CONSTRUCTION	METHOD	WHAT IT ALSO GIVES
perpendicular bisector of AB	equal arcs from A and from B, joined	the midpoint of AB
perpendicular from P to a line	an arc from P cutting the line twice, then the bisector of that segment	the distance from P to the line
bisector of an angle	equal arcs from the vertex, then from the two crossings	the locus of equidistant points
an angle of $60°$	an equilateral triangle on a segment	$30°$

WHY THE PERPENDICULAR BISECTOR WORKS

Both arc centres are the same distance from the two crossing points, so each crossing is equidistant from A and B. The line through them is the set of all such points — and that line meets AB at right angles, at its midpoint.

02 Worked examples

EXAMPLE 1 **Construct a triangle with sides $6\ cm$, $5\ cm$ and $4\ cm$.**

1. Check: $4 + 5 > 6$ ✓

   The longest side only.

2. Draw $BC = 6\ cm$.

3. Arc of $5\ cm$ from B, arc of $4\ cm$ from C.

4. Join the crossing to both ends.

   Congruent by SSS to any other such triangle.

ANSWER

Constructed

EXAMPLE 2 **Can a triangle with sides *5, 8, 14* be constructed?**

1. The longest side is 14.
2. $5 + 8 = 13 < 14$
3. The two arcs never meet.
4. No such triangle exists.

ANSWER

No

EXAMPLE 3 **Construct the midpoint of a segment *AB*.**

1. With centre A and radius more than half AB, draw an arc.
2. Repeat from B with the same radius.
3. Join the two crossings.
4. That line meets AB at its midpoint, at right angles.

ANSWER

The perpendicular bisector gives it

03 Interactive model

Set 5, 8 and 14 cm without comment; the class works for two minutes before discovering the arcs cannot meet, and the triangle inequality is never forgotten again.

Which construction, and does it work?

The two allowed tools are:

- ruler and protractor
- straightedge and compasses
- compasses and protractor
- ruler and set square

Arcs should be:

- rubbed out
- left on the page
- drawn in pen
- omitted

Three sides fix a triangle by:

- the first criterion
- the second
- the third
- none

Sides $5, 8, 14$ can be constructed:

- yes
- no
- only with a protractor
- sometimes

Two sides and the angle between them use:

- the first criterion
- the second
- the third
- none

The perpendicular bisector also gives:

- the midpoint
- the bisector of an angle
- the area
- the centroid

An equilateral triangle constructs an angle of:

- $45°$
- $60°$
- $90°$
- $30°$ directly

Every point of the perpendicular bisector is:

- on the segment
- equidistant from the ends
- the midpoint
- outside

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Construction	Yasash	Построение
Compasses	Sirkul	Циркуль
Straightedge	Chizg‘ich	Линейка
Arc	Yoy	Дуга
Perpendicular bisector	O‘rta perpendikulyar	Серединный перпендикуляр
Midpoint	O‘rta nuqta	Середина
Given data	Berilganlar	Данные
Impossible	Yasab bo‘lmaydi	Невозможно

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

Compasses may be used to:

- measure an angle
- draw a circle of given radius
- draw a straight line
- estimate

The construction from three sides fails when:

- the sides are equal
- the triangle inequality fails
- one side is long
- never

A side and the two angles on it use:

- the first criterion
- the second criterion
- the third criterion
- none

The perpendicular bisector of AB passes through:

- A
- the midpoint of AB
- B
- no special point

Construction arcs should be:

- erased
- left visible
- coloured
- redrawn

A construction is justified by:

- measuring the result
- a congruence criterion
- the drawing looking right
- nothing

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. The two allowed tools

   ANSWER straightedge and compasses

2. Three sides fix a triangle by

   ANSWER the third criterion

3. Two sides and the included angle

   ANSWER the first criterion

4. A side and two angles on it

   ANSWER the second criterion

5. Sides $6, 5, 4$: constructible?

   ANSWER Yes

6. Sides $5, 8, 14$: constructible?

   ANSWER No

7. The perpendicular bisector gives the

   ANSWER midpoint

Medium · the standard question

7 PROBLEMS

1. Sides 3, 4, 8: constructible?

ANSWER No

2. Sides 7, 7, 13: constructible?

ANSWER Yes

3. Which construction gives an angle of $60°$?

ANSWER An equilateral triangle

4. And of $30°$?

ANSWER Bisect the $60°$

5. And of $90°$?

ANSWER A perpendicular to a line at a point

6. And of $45°$?

ANSWER Bisect the $90°$

7. Why leave the arcs?

ANSWER They show the construction was not estimated

Hard · stretch and reason

7 PROBLEMS

1. Construct a triangle from two sides 5 and 7 with the angle between them $60°$

   ANSWER Copy the angle, mark the lengths on its arms

2. How many triangles arise from three given sides?

   ANSWER One, up to congruence

3. Construct an angle of $15°$

   ANSWER Bisect $60°$ twice

4. Construct the perpendicular from a point to a line

   ANSWER An arc cutting the line twice, then bisect that segment

5. Why does the perpendicular-bisector construction work?

   ANSWER Both crossings are equidistant from the two ends

6. Given the perimeter and two angles, is the triangle fixed?

   ANSWER Yes — the third angle follows and the shape scales to the perimeter

7. Sides $2, 3, 5$: constructible?

   ANSWER No — the points would be in a line

07 Homework

Homework — 5 tasks

Check the triangle inequality before touching the compasses.

1. Construct a triangle with sides $7\ cm$, $5\ cm$ and $4\ cm$.
2. Try to construct one with sides $3\ cm$, $4\ cm$ and $9\ cm$, and explain what happens.
3. Construct an angle of $60°$ and bisect it.
4. Construct the perpendicular bisector of a $8\ cm$ segment and mark its midpoint.
5. Name the congruence criterion that justifies each of the three triangle constructions.